A reconnaissance paleomagnetic survey of the central Labrador Trough, Quebec[1]

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Paleomagnetic studies have been carried out on lower Proterozoic (Aphebian) rocks of the Labrador Trough in central Quebec. The magnetizations are complex. Most magnetizations are secondary and none are demonstrably original, but their relative order can be estimated from their associated blocking temperatures. The oldest magnetizations (*A*) are thought to have been acquired at about 1800 Ma following deposition but not necessarily long afterwards. These magnetizations have blocking temperatures of about 600 °C and occur in the Seward Subgroup (mainly arkoses), the Menihek Formation (argillites), and the Retty Formation (peridotites). The Castignon carbonatitic complex also has *A* type magnetizations which are referred, tentatively, to a K–Ar date of 1875 ± 50 Ma. The next youngest magnetizations (*B*) have blocking temperatures commonly distributed between 300 and 500 °C and are considered to have been acquired at about 1600 Ma during uplift following the Hudsonian orogeny. *B* magnetizations are the commonest magnetizations and are found in the Seward, the Retty, and the Menihek. The youngest magnetizations (*C*) have low blocking temperatures predominantly less than 300 °C and were probably acquired long after the rocks were formed. The *C* magnetizations are found in the Seward Subgroup, the Willbob Formation (basalts), and the Wakuach Formation (gabbros). The age of the *C* magnetization appears to be about 1100 Ma, but its origin is not known. This chronology of magnetizations is consistent with the previously constructed path of apparent polar wandering for Laurentia.

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Geology and Geochronology

The Labrador Trough, together with the Cape Smith Belt and the Belcher Islands Belt, is an erosional remnant of a possibly once continuous belt of Aphebian rocks called the Circum-Ungava Geosyncline (Fig. 1). The present paleomagnetic study is the result of a reconnaissance survey carried out in the central Labrador Trough, where no previous paleomagnetic work had been done. Geological mapping and interpretation in the sampling area has been carried out by Fahrig (1956, 1957), Roscoe (1957), Baragar (1967), and Dimroth (1968*a*, 1968*b*, 1970*a*, 1970*b*, 1971, 1972), and reviewed by Dimroth *et al.* (1970).

The general stratigraphic succession is set out in Table 1. The evolution of the Labrador Trough can be divided into two and possibly three tectonic cycles (Fahrig 1957; Dimroth 1968*b*; Dimroth *et al.* 1970), each of which began with the deposition of orthoquartzites and chemical sediments (*e.g.* Seward, Sokoman) and ended with the deposition of shales and quartzites (*e.g.* Menihek) in the miogeosyncline, and volcanics (*e.g.* Willbob, Wakuach, Retty) in the eugeosyncline (Fig. 1). The three cycles are defined by the rock units Seward to Attikamagen, Wishart to Murdoch, and Thompson to Retty (Table 1). The first cycle began with the deposition on Archean terrain of the sandstones and dolomites of the Seward Subgroup. These were collected at eight sites from mainly fine- and medium-grained red arkoses (Fig. 2). The Sokoman Iron Formation, which was deposited near the beginning of the second cycle, was sampled at four sites (Fig. 2). The immediately overlying Menihek Formation (in the western Trough) was sampled from six sites of very fine-grained, maroon to almost black, argillaceous rocks (Fig. 2). A possible third cycle ended with the extrusion of the Willbob basalt. The Wakuach gabbro and Retty peridotite are probably in large part equivalent to the Willbob, but may include sills equivalent to the Menihek and earlier volcanics. Samples were collected from 11 sites in the Willbob, 9 in the Wakuach, and 3 in the Retty (Fig. 2). The units from the Seward Subgroup to the Willbob Formation compose the Kaniapiskau Supergroup (Frarey and Duffell 1964). The last unit to be sampled was the Castignon complex which comprises scattered outcrops of carbonatitic rocks. Five sites were sampled in the carbonatitic intrusions west of Castignon Lake (Fig. 2); three sites from a dia-

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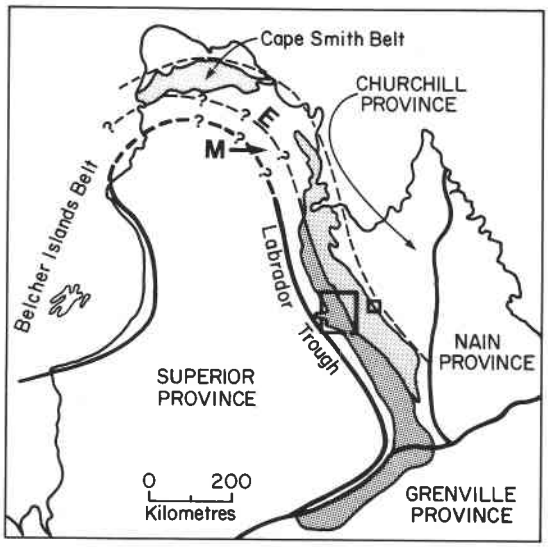


FIG. 1. Generalized location map after Stockwell (1964) and Dimroth (1970*b*). Sampling areas are indicated by rectangles. Broken lines indicate the probable former limits of and boundaries between the general lithologies of the Circum-Ungava Geosyncline (Belcher Islands Belt, Cape Smith Belt, Labrador Trough). Heavy lines separate the Structural Provinces. M stands for miogeosyncline, E for eugeosyncline.

TABLE 1. Stratigraphic sequence

Montagnais Group
Retty Formation*
Wakuach Formation*
Doublet Group
Willbob Formation*
Thompson Lake Formation
Murdoch Formation
Knob Lake Group
Menihek Formation*
Castignon complex (?)*
Sokoman Formation*
Ruth Formation
Wishart Formation
Attikamagen Formation
Swampy Bay Subgroup
Pistolet Subgroup
Seward Subgroup*

*Rock units sampled.

treme (carbonatite tuff breccia and flow-layered carbonatite), and two sites from separate dikes (carbonatitized meimechite). The Castignon complex was first considered to be post-orogenic (Dimroth 1970*a*), but Dressler (1975) has recently argued that it is pre-Menihek and of probable Upper Sokoman age.

Baragar (1967) considered the folding and faulting that occurred during the period corresponding to the first two cycles as part of a continuous period of deposition and deformation, which he called the 'geosynclinal phase'. According to Baragar this activity culminated in the major period of deformation or 'orogenic phase' known as the Hudsonian Orogeny. Indeed, the voluminous magmatism represented by the Willbob, Wakuach, and Retty rocks could have heralded the commencement of this Orogeny. During the orogeny the Trough was intensely folded and faulted, with overthrusting directed westward. Post-Hudsonian activity in the Cape Smith Belt and Labrador Trough included several intrusive episodes, notably the Montagnais dikes to the south of the sampling area. Cretaceous nonmarine clastics are recorded from near Schefferville, and their deposition was accompanied by some deformation and faulting (Stubbins *et al.* 1961; Bergeron 1965). Metamorphism in the Trough increases generally from west to east, ranging in the sampling area from the pumpellyite facies (Seward Subgroup and Menihek Formation), through the quartz–albite–muscovite–chlorite subfacies of the greenschist facies (Willbob basalt, Wakuach gabbro) to the almandine–amphibolite facies (Retty peridotite). High temperatures were apparently maintained in the central region for some time after the Hudsonian deformation.

A number of radiometric dates are available from the area. Rocks of the Trough and its basement, reworked in the Hudsonian orogeny, yield ages that range from 1575 to 1935 Ma (13 determinations, Beall *et al.* 1963, in Dimroth 1970*b*). Diabase dikes, which intrude the basement but not the supracrustal rocks, have yielded K–Ar dates of about 2150 Ma which apparently set a lower age limit to the rocks of the Trough. Trough rocks have been dated by K–Ar, Rb–Sr, and lead methods. K–Ar ages range between 2060 and 1800 Ma (Beall *et al.* 1963), and between 1555 and 1690 Ma. Fryer (1972) considered the 2060 Ma date as being too old due to an excess of ^{40}Ar. The latter K–Ar ages have an average of 1610 Ma (11 determinations) and probably date the later stages of post-Hudsonian uplift (Dimroth 1970*b*, Fryer 1972). Rb–Sr whole-rock isochrons determined by Fryer (1972) are interpreted by him as primary ages. He derives an age of 1870 $\pm$ 50 Ma (decay constant = 1.39×10^{-11} yr^{-1}) for the Sokoman Iron Formation, which is supported by an 1855 $\pm$ 74 Ma age for the overlying Menihek Formation,

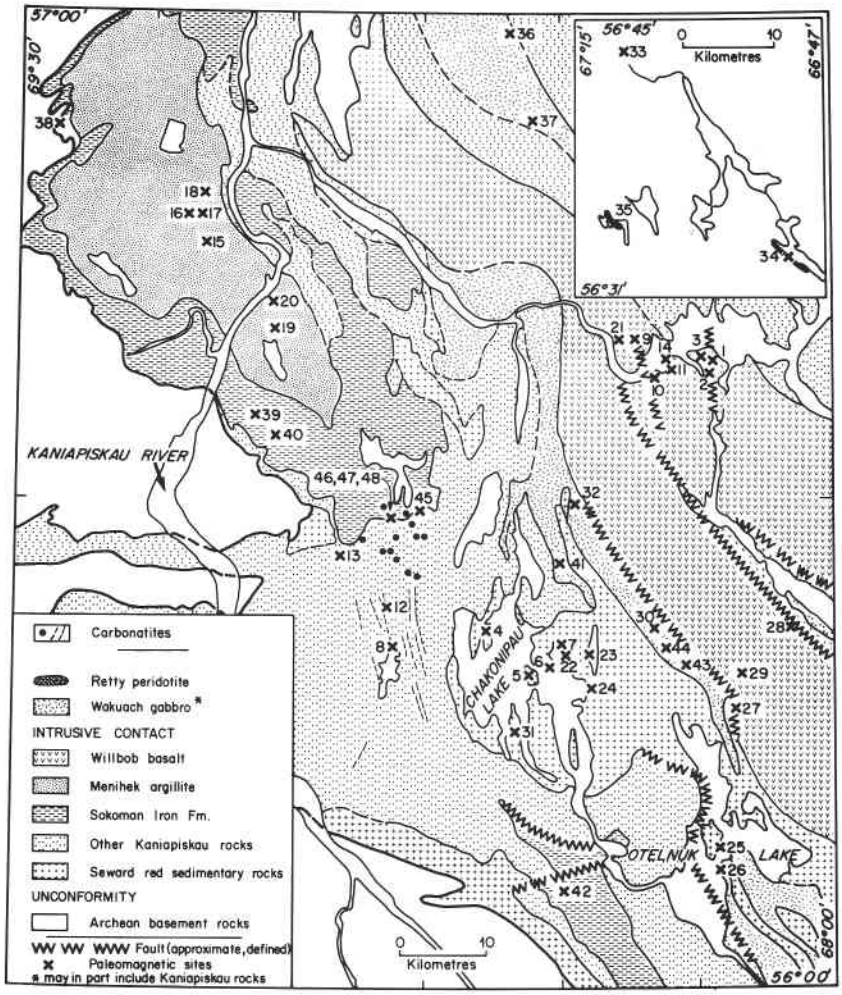


FIG. 2. Sites and generalized geology (after Fahrig 1956, Roscoe 1957, and Dimroth 1968*b*).

and a 1900 Ma reference isochron for the stratigraphically older Attikamagen Formation. Lead isotope ages of 1800 Ma (Pb206/204) and 1660 Ma (Pb208/204) quoted in Roscoe (1957) have been obtained from galenas found in stromatolitic dolomite of the Alder Formation of the Pistolet Subgroup (Table 1). The former date may relate to primary deposition, and the latter to post-Hudsonian uplift. Finally, Dressler (1975) has reported a K–Ar date of 1875 ± 50 Ma from a lamprophyre pipe of the Castignon complex, which he considers to be contemporaneous with the upper Sokoman.

Methods

Sampling was carried out by helicopter in collaboration with the Gravity and Geodynamics Division of the Earth Physics Branch. Most samples were obtained with a portable drill, usually from hilltops or lake shores; three to five cores were drilled at each site and oriented *in situ* by sun and magnetic compasses. Two specimens were generally cut from each core. In the laboratory, specimens were demagnetized thermally and by alternating fields (af) in the apparatus described by Roy *et al.* (1969, 1971, 1973), and measured on a Schonstedt spinner magnetometer (SMI). The fields and temperatures used for magnetic and thermal cleaning were determined from a combination of the end-point (As and Zijderveld 1958) and least scatter (Irving *et al.* 1961) methods. Sample directions (the mean of results from two specimens) were used as the basic statistical units.

Results from samples were rejected (1) if the difference between the directions in specimen pairs exceeded 50°, (2) if the specimen directions exhibited erratic behaviour during demagnetization due probably to the acquisition of spurious magnetizations during the measurement or demagnetization procedures, or (3) if the intensities were less than 3×10^{-4} A/m (1 A/m $= 10^{-3}$ emu) which we find to be the lower limit of reliability for this magnetometer. Site results were rejected if they had random directions (Watson 1956*b*; Vincenz and Bruckshaw 1960). All directions are referred to the present horizontal unless otherwise indicated.

Vector analysis was used in some sites to separate magnetizations whose remanent coercive force or blocking temperature spectra, although they may overlap, are nevertheless clearly different (Roy and Park 1974). An example is given in Table 2. Here the natural remanent magnetization (nrm) (direction: 150°, −71°) is composed of two magnetizations: *A* with a direction of approximately 192°, +05° and *B* with a direction of approximately 128°, −76°. Much of the blocking temperature spectrum of *A* resides in the 550 to 650 °C interval; the fact that the direction remains constant within that interval indicates that *A* has been successfully isolated and that the blocking temperatures of *B* lie between room temperature and 550 °C. The gradual increase of declination and inclination (less negative) between 20 and 550 °C is attributable to the decay of *B* and perhaps also to the decay of *A*. It is therefore possible that the blocking temperature spectra of *A* and *B* overlap. However, even if this is so, the fact that *A* has been isolated means that both *A* and *B* can be determined by vector analysis and the mean values of the 550 and 600 °C results (Table 2) can be regarded as good estimates of the *A* (column 1) and *B* (column 2) vectors. The result at 650 °C is less reliable, because, since only a small portion of *A* remains, the accuracy of determination is reduced.

To determine if anisotropy has affected the remanence directions, the test of Irving and Park (1973) was applied. Specimens were oriented between the poles of an electromagnet in their *in situ* positions and an isothermal remanence (irm) induced by applying a strong magnetic field. They were then cleaned in alternating fields. Differences between the direction of the applied field and the direction of irm would be due to anisotropy. It is evident from Table 3 that there were no significant differences; therefore the effect of anisotropy is negligible.

TABLE 2. Example of vector subtraction from site 23

T (°C)	(1) *M*	(1) *D*°,*I*°	(2) *M*	(2) *D*°,*I*°
Nrm	71.6	150,−71	—	—
100	66.3	148,−69	5.9	268,−81
200	37.8	161,−64	34.6	126,−77
300	25.0	168,−56	48.2	127,−77
400	16.7	177,−44	57.8	127,−76
520	11.1	183,−19	66.0	128,−76
550	9.6	192,+00	69.9	128,−76
600	9.7	192,+05	70.7	128,−76
650	1.9	195,+07	71.4	146,−72

NOTES: The vector differences (2) are the measured values at the temperatures indicated subtracted from the natural remanent magnetization given in column (1). *M* is the intensity in 10^{-3} A/m. *D*°,*I*° are the declination and inclination.

Seward and Menihek Formations

Natural remanent magnetization (nrm) directions were commonly well-grouped and southerly (Fig. 3, site 6), with Fisher's statistic *k* ranging from 30 to 700 in the Seward, and from 2 to 65 in the Menihek. Three sites are reversed (upward directed nrms, Fig. 3); one having a very high precision (Fig. 3; site 23, $k = 700$). During thermal cleaning most site directions moved away from the earth's field to a southerly direction of generally shallower inclination. Thermal demagnetization data are depicted in Fig. 4, which shows the standard deviation envelopes of the mean curves. The blocking temperatures are distributed over a wide range, and are commonly less than 300 °C (Fig. 5*b*, *c*). Seward

TABLE 3. Effect of anisotropy

	D°,*I*°	(*N*,*k*, α_{95}°)
Willbob and Wakuach		
Applied field (1.9 T)	307,+67	
Irm before af cleaning	305,+66	(8,466,3)
Irm after af cleaning	299,+65	(8,338,3)
Retty		
Applied field (1.9 T)	193,+00	
Irm before af cleaning	194,−03	(2,1300,7)
Irm after af cleaning	194,−03	(2,524,11)
Castignon Complex		
Applied field (1.9 T)	216,+67	
Irm before af cleaning	214,+69	(3,615,5)
Irm after af cleaning	220,+67	(3,1430,3)

NOTES: *N* is the number of specimens, *k* the precision, and α_{95}° the 95% circle of confidence in degrees. Other symbols are explained in Table 2.

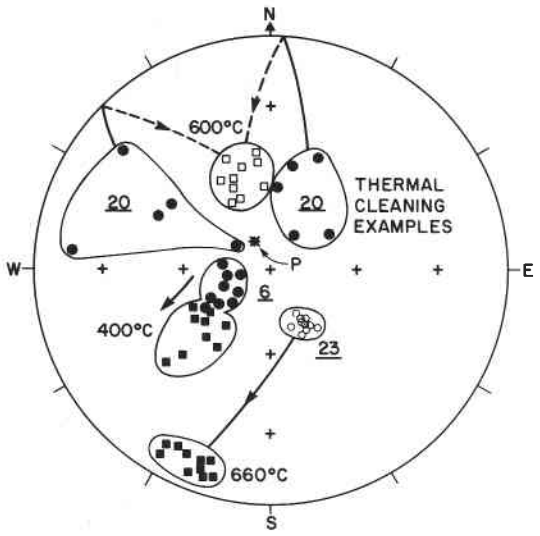


FIG. 3. Examples of nrm and thermal cleaning of Seward and Menihek sites. P represents the direction of the present earth's field. Directions are plotted with respect to present horizontal. Solid (open) symbols in this and succeeding figures are plotted on the lower (upper) hemisphere. Circular (square) symbols are nrm (thermally cleaned) directions.

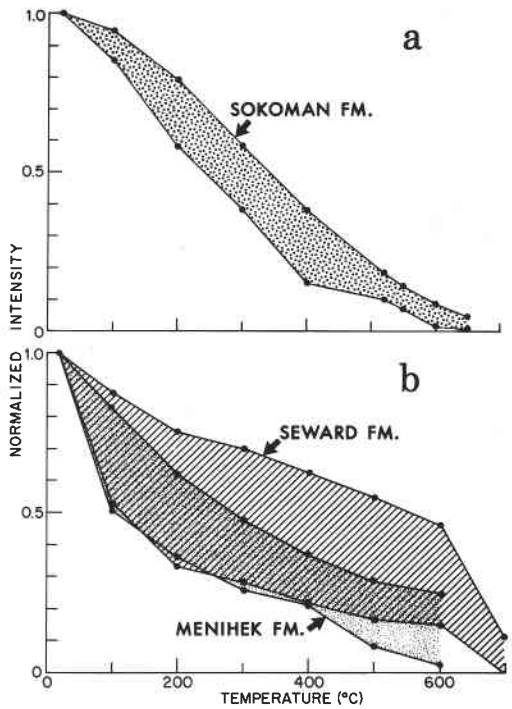


FIG. 4. Summary of thermal demagnetization curves of specimens from the metasedimentary rock units. Shaded areas are the standard deviation about the mean curves (not shown) for (*a*) the Sokoman Iron Formation (8 specimens, 4 sites) and (*b*) the Seward Subgroup (16 specimens, 8 sites) and Menihek (12 specimens, 6 sites) Formation.

rocks display a noticeable range of blocking temperatures between 600 and 700 °C (Fig. 5), but most samples from the Seward and Menihek have random directions in this high temperature range. Magnetizations with such a distribution of blocking temperatures are readily reset by mild heating during burial (Irving and Opdyke 1965; Pullaiah *et al.* 1975). The cleaned sample directions for the two rock units are depicted in Figs. 6*a* and 7*a*. Individual site directions are given in Table 4, and their averages in Table 5 (entries 4 and 5).

Vector analysis was used to separate the magnetizations present. The magnetizations of specimens after cleaning in the six very stable Seward sites were subtracted vectorially from the nrm vectors (see Table 2 for an example of the procedure). Site results are given in Table 6, and the directions of the removed magnetizations, which are presumably overprints, are plotted in Fig. 6*b*. The mean of the upward directions at site 23 agrees well, irrespective of sign, with the mean direction of the five downward-directed sites, and the overall mean (Table 5, entry 3) is significantly different from the direction of the present earth's field, and that of the geocentric axial dipole field. The good agreement between normal and reversed directions as well as the high precision of the directions indicate that this 'overprint' has been clearly separated. Vector subtraction was also applied to those Menihek samples which appeared also to have a substantial magnetic overprint. The directions of their removed magnetization (Fig. 7*b*) were more scattered, although the mean direction is close to that derived from the Seward sites (Table 5, entry 2).

The very stable magnetizations are associated with hematite, the unstable (site 24) with magnetite. In the Menihek 50% of the opaque minerals is goethite. Goethite is capable of carrying a weak thermoremanent magnetization at temperatures below 350 °C (Strangway *et al.* 1968), and in the present study may account for some of the less-stable magnetization in the Menihek below 300 °C. However, the goethite is not considered to be carrying much remanence in these rocks, because the Seward, which has a similar low range of blocking temperatures to the Menihek, contains no goethite.

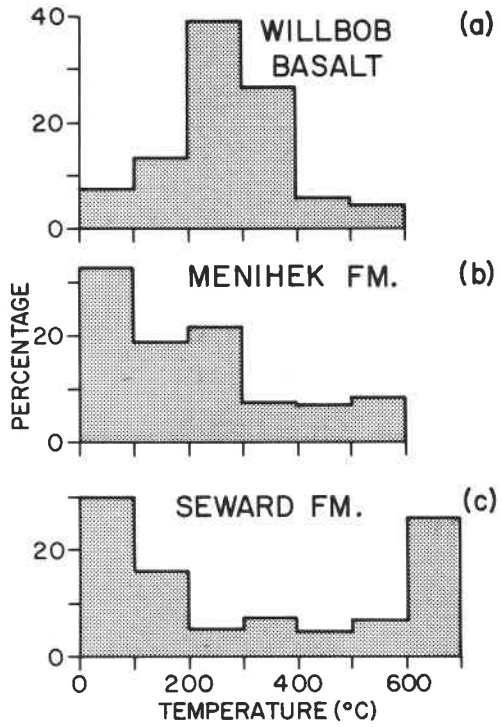


FIG. 5. Blocking temperature spectra of the Willbob (a), Menihek (b), and Seward (c) rocks. The spectra represent averages obtained from 11, 12, and 16 specimens respectively. The missing percentages in these histograms are noise components beyond the highest blocking temperatures depicted.

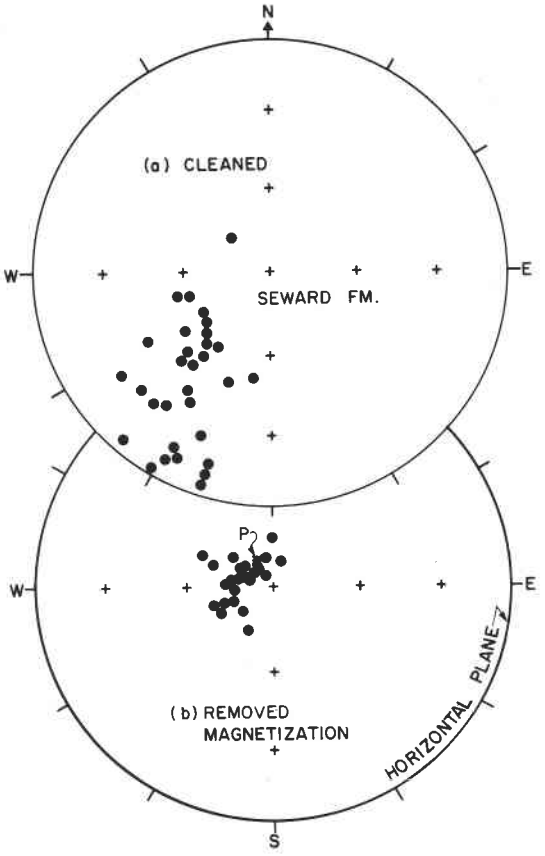


FIG. 6. Sample directions from the Seward Subgroup. (*a*) Final cleaned directions. (*b*) Directions obtained by vectorially subtracting the final cleaned magnetization from nrm (see example Table 2). Removed magnetization at site 23 has been reversed for ease of comparison.

Each Seward and Menihek site generally yields two different directions. The different magnetic phases are indexed by the letters *A*, *B*, and *C* denoting a possible age sequence. The direction obtained by vector analysis (Table 6) appears to be associated with that magnetization having blocking temperatures mainly less than 300 °C. This magnetization (*SC* or *MC*) is probably a viscous partial thermoremanent magnetization (ptrm). The *A* and *B* magnetizations with blocking temperatures up to 670 °C (Table 4) are carried by hematite. They could be thermoremanent or chemical remanent (crm), and either of them could be primary or both could be secondary (Roy and Lapointe 1976, see p. 764). The large scatter of the cleaned sample directions (Figs. 6*a* and 7*a*) could be due to unresolved magnetizations, that is, *A* and *B* mixed together. The inclinations vary from steep to shallow, and the directions have been separated (though somewhat arbitrarily) into a main group with steep inclinations (*SB* and *MB*), and a subsidiary group with low inclinations (*SA* and *MA*).

There is some question as to whether these directions should be corrected for bedding. The steep Seward and Menihek directions (main groups) were therefore averaged before and after correction for bedding. As shown in Table 5 the precision of the Menihek sites improves slightly when corrected for bedding (compare first and third lines of entry 4), but this improvement is not significant at the 95% confidence level (Watson 1956*a*). The precision of the Seward sites, however, does not improve but worsens (Table 5, compare first and third lines of entry 5), and the directions with respect to horizontal have a better grouping. The change is significant. The same conclusions can also be drawn when the sites of low and steep inclination

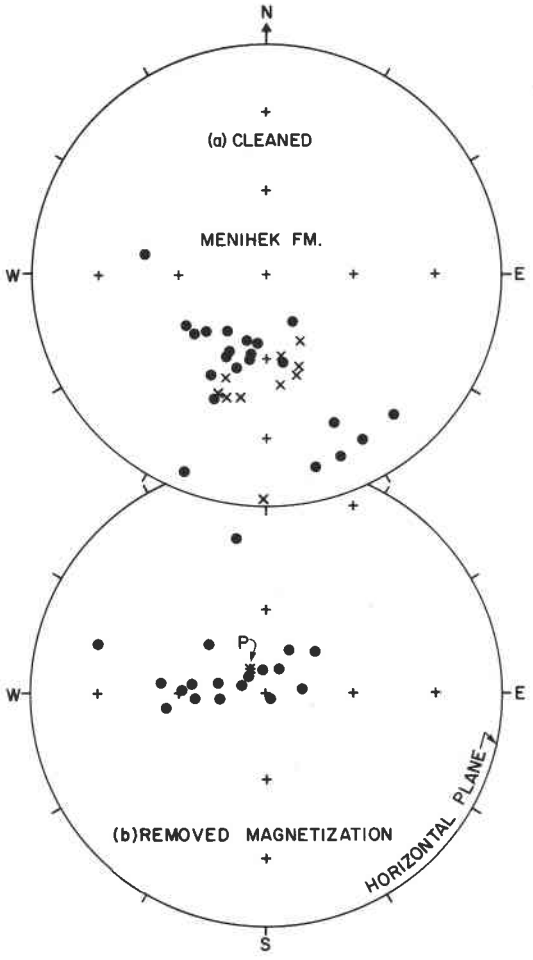


FIG. 7. Sample directions from the Menihek Formation. (*a*) and (*b*) are as explained in Fig. 6. Directions denoted by crosses on the lower hemisphere have been reversed 180° from their actual positions.

are not separated and all the sites are considered together. Therefore the correct direction for both rock units is considered to be that given with respect to horizontal. Since the Seward and Menihek are the least metamorphosed rocks of the Trough, it is reasonable to suppose that the directions of other rock units should likewise be referred to the horizontal plane. This implies that all magnetizations were acquired after the main deformational phase.

Willbob Basalt, Wakuach Gabbro, and Retty Peridotite

The nrm directions at most sites, though often scattered (site precisions (k) ranging from 1 to 40), increase in precision during af demagnetization. Many samples have af decay curves like those from sites 10, 14, and 33 depicted in Fig. 8, and their directions become random in low fields. The Willbob basalt is the most unstable of the three rock units: out of the 64 samples collected 53 were rejected on the criteria given above, compared to 8 rejected from the 33 Wakuach samples, and 4 from the 15 Retty samples. Low cleaning fields (typically 10 to 20 mT (1 mT = 10 gauss)) and low median destructive fields (mdfs) of nrm (typically five times less than the mdfs of irm, Table 7) for these rock units indicate that a large viscous component is probably present. Final directions of the more stable of the Willbob and Wakuach sites are broadly similar (Table 8), and they have therefore been averaged together in Table 5 (entry 1) and plotted together in Fig. 9*a*. The Willbob-Wakuach direction (*WC*) is similar to the *SC* and *MC* directions in Seward and Menihek rocks, but has a larger scatter (compare Figs. 9*a*, 6*b*, 7*b*). The Retty peridotite yielded two distinct southerly directions with high and low inclinations (*RB* and *RA* Table 8, and Table 5 entries 6 and 10; Fig. 9*b*).

Thermal demagnetization curves from stable Willbob sites show a large intensity drop between 300 and 400 °C (Fig. 10*a*; sites 2 and 9). Curves from unstable sites lie between the curves of sites 10 and 14. Directions are usually only stable up to 200 or 300 °C (9 of 13 sites), but occasionally are retained up to near the Curie point of magnetite. Blocking temperatures are low (Fig. 5*a*), and probably mainly due to pyrrhotite, the dominant opaque mineral in polished sections. Some of the low blocking temperature remanence may be contributed by goethite which is also present.

The thermal behaviour and the opaque minerals of the Wakuach gabbro are different from those of the Willbob basalt. Thermal decay curves show higher blocking temperatures with distinct ranges indicative of magnetite (*e.g.* Fig. 10*b*, sites 30 and 43) and of hematite (*e.g.* Fig. 10*b*, sites 25 and 30). Directions usually scatter between 500 and 600 °C near the Curie point of magnetite, but samples from five sites each moved to another direction between 600 and 735 °C (discussed below). The opaque minerals are dominated by magnetite or titanomagnetite, which is often heavily granulated and as much as 50% altered to goethite. Some hematite also occurs, mainly as blebs, lamellae, and patches

TABLE 4. Thermally cleaned directions of metasedimentary rocks

Site	UTMG	M_N	T(°C)	N	$D°,I°(R)$
		Seward Subgroup			
4	EN217466	22.7	670	3(2)	254,+56(2.77)
5	EN270417	4.6	550	3(2)	205,+21(2.77)
6	EN295419	2.2	400	5	226,+62(4.95)
7	EN306446	14.1	660	5	220,+30(4.94)
22	EN310437	9.4	660	5	222,+51(4.98)
23	EN342434	41.9	660	5	202,+10(4.98)
24	EN341398	1.7	600	(5)	—
31	EN255348	14.9	660	4(1)	221,+32(3.69)
		Menihek Formation			
15	DP907909	69.8	600	5	206,+64(4.96)
16	DP890942	16.9	550	5	230,+56(4.77)
17	DP898942	41.6	660	5	014,−37(4.75)
18	DP904961	12.2	600	5	154,+21(4.92)
19	DP982810	89.7	600	5	184,+62(4.93)
20	DP979843	27.4	600	5	345,−56(4.97)
		Sokoman Iron Formation			
38	DP737042	2659	550	4	053,+48(3.26)
39	DN968702	10 440	550	4	003,+56(3.79)
40	DN982687	4138	550	3	096,+65(2.63)
42	DN313167	4091	550	4	219,+07(2.97)

NOTES: UTMG is the universal transverse mercator grid reference. All refer to grid zone 19V. M_N is the nrm intensity in 10^{-3} A/m, T (°C) is the thermal cleaning temperature in °C, N is the number of samples (cores) with those in parentheses discarded, $D°,I°$ are the declination and inclination, and R is the vector resultant.

TABLE 5. Summary of directions and poles

Entry	Description	N	$D°,I°$	k	$\alpha_{95}°$	Pole ($dm°,dp°$)
1. WC	Wakuach–Willbob directions	9	305,+67	11	16	54N,146W(27,22)
—	Wakuach–Willbob poles	9	—	6	24	61N,140W
2. MC	Menihek directions	4	298,+72	32	16	57N,132W(29,26)
—	Menihek poles	4	—	14	26	59N,131W
3. SC	Seward directions	6	281,+77	175	5	53N,112W(09,09)
—	Seward poles	6	—	53	9	53N,112W
4. MB	Menihek directions*	5	195,+57	23	16	05N,081W(24,17)
—	Menihek poles*	5	—	14	21	06N,081W
—	Menihek directions* (bedding)	5	193,+58	41	12	—
5. SB	Seward directions†	5	227,+47	23	17	03N,109W(21,14)
—	Seward poles†	5	—	23	16	05N,109W
—	Seward directions† (bedding)	5	225,+39	3	49	—
6. RB	Retty direction, site 34(*i*)	1	199,+56	—	—	05N,082W
7. CB	Castignon complex directions	5	217,+67	69	9	21N,093W(15,13)
—	Castignon complex poles	5	—	32	14	22N,092W
8. MA	Menihek direction, site 18	1	154,+21	—	—	19S,042W
9. SA	Seward direction, site 5	1	205,+21	—	—	20S,095W
SA	Seward direction, site 23	1	202,+10	—	—	26S,093W
10. RA	Retty direction, site 34(*ii*)	1	183,+10	—	—	28S,070W
RA	Retty direction, site 35	1	193,+04	—	—	31S,082W

NOTES: N is the number of sites. $dm°,dp°$ are the semi-axes of the polar error. *WC*, *MC*, *SC*, and *RB* (see text) were obtained by vector subtraction. Other symbols are explained in Tables 2 and 3. Unless otherwise stated, the *in situ* directions are listed.
*Excluding site 18 (see entry 8).
†Excluding sites 5 and 23 (see entry 9).

TABLE 6. Mean directions of the magnetizations removed by thermal cleaning

Site	N	$D°,I°(R)$
	Seward Formation	
4	(4)	—
5	5	294,+79(4.98)
6	5	302,+80(4.95)
7	5	246,+76(4.96)
22	5	265,+74(4.97)
23	5	102,−72(4.99)
24	Unstable	
31	5	306,+76(4.92)
	Menihek Formation	
15	(5)	—
16	5	289,+79(4.91)
17	3(2)	270,+56(2.99)
18	5	336,+73(4.55)
19	5	Random(2.86)
20	5	319,+72(4.42)

NOTES: The directions were obtained by subtracting vectors obtained after thermal cleaning from nrm vectors. Symbols are explained in Tables 2 and 4. The number of samples not used (those with a large component parallel to the present field) is noted in parentheses. Sample directions are plotted in Figs. 6 and 7.

in ilmenite.The geothite may carry some of the remanence below 350 °C, but it would be a very small proportion considering the intensity of these rocks.

Thermal decay curves of the Retty peridotite show blocking temperatures between 500 and 600 °C characteristic of magnetite (Fig. 10*c*). The directions become random above this range. Magnetite indeed occurs, along with ilmenite, sulfides, the occasional hematite, and the occasionally dominant pyrrhotite.

The stable magnetization in the Willbob and Wakuach rocks (*WC*; Table 8) is probably due, in part, to magnetite. It has a similar direction to the low blocking temperature magnetization of the Seward and Menihek metasediments (*SC* and *MC*), and is thus possibly a related magnetization. Most Wakuach samples are located within fault zones and have higher blocking temperatures (500–600 °C) than the Willbob samples (200–400 °C). It is therefore possible that the Wakuach magnetization is largely a crm resulting from chemical changes in the iron minerals close to the fault zones, and that the magnetization of the Willbob basalt and the low blocking temperature magnetization in the metasediments is a viscous ptrm related in some way to a contemporaneous heating event.

Neither of the directions of magnetization of

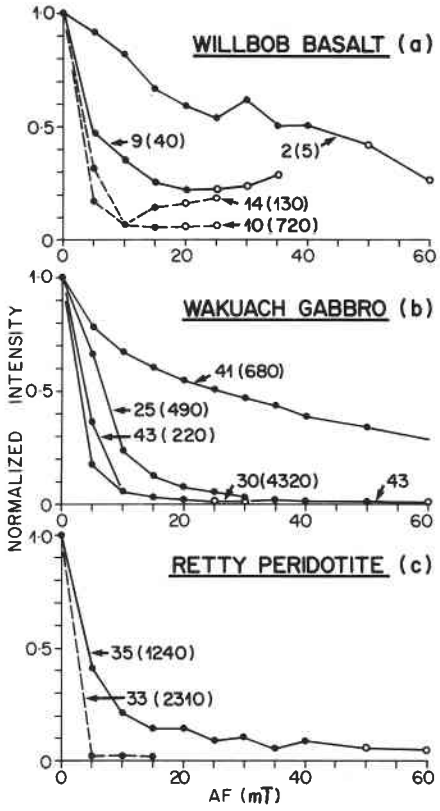


FIG. 8. Typical af decay curves (labelled by site numbers) of Willbob (*a*), Wakuach (*b*), and Retty (*c*) rocks. Unstable specimens are shown by dashed curves. Numbers in brackets are the nrm intensities in 10^{-3} A/m. Solid dots (open circles) denote directions that are less (more) than about 20° from the cleaned direction.

TABLE 7. Median destructive fields (mT)

Rock unit	n	Nrm	Irm
Willbob basalt	11	7.1±9.4	33.0±17.2
Wakuach gabbro	8	5.6±5.0	20.2±11.3
Retty peridotite	3	2.7±1.6	17.7±6.7
Carbonatite	4	5.3±1.3	22.1±4.5

NOTES: n is the number of specimens. The irm (isothermal remanent magnetization) was acquired in 1T.

the Retty peridotite (*RA* and *RB*) agrees with the direction in the Willbob and Wakuach rocks. The less-stable of the Retty magnetizations (*RB* at site 34*i* in Table 8) has a direction close to that of the *SB* and *MB* magnetizations of the metasedimentary rocks and could therefore be a viscous ptrm associated with magnetite. The more stable of the Retty magnetizations (*RA* at sites 34*ii* and 35 in Table 8) could be a crm

TABLE 8. Cleaned directions of magmatic rocks

Site	UTMG	M_N	Treatment (mT)	N	$D°,I°(R)$
1	EN478774	275	10	3(2)	248,+46(2.77)
2	EN472758	12	15	4(1)	292,+29(3.84)
3	EN462775	127	15	2(3)	283,+39(1.65)
9	EN387795	23	15	3(2)	196,+71(2.88)
28	EN567468	63	15	3(2)	Random(2.62)
29	EN511420	100	15	3(2)	291,+60(2.80)
		Wakuach gabbro			
25	EN487216	1164	20	5	312,+51(4.63)
30	EN414464	4427	15	5	317,+71(4.90)
32	EN322608	1895	15	5	030,+79(4.70)
37	EP2805	683	15	2(3)	334,+53(0.96)
41	EN309541	388	100	5	305,+45(4.87)
43	EN452428	53	15	3(2)	303,+72(2.87)
44	EN425445	2930	15	3	030,+72(2.66)
		Retty peridotite			
33	FN1487	2127	15	4(1)	Random(2.33)
34(*i*)	FN3264	425	15	5	199,+56(4.70)
34(*ii*)	FN3264	—	550*	5	183,+10(4.63)
35	FN1267	3126	550*	2(3)	193,+04(2.00)
		Miscellaneous gabbro			
13	EN059550	2413	15	4(1)	033,+83(3.69)

NOTES: Symbols are explained in Tables 2 and 4. Willbob (10, 11, 14, 21, 27) and Wakuach (26) sites with one or no stable samples are not listed. Sites 1, 3, and 37 show erratic behaviour during af demagnetization and are not used in the analysis. Site 25 may belong to the Kaniapiskau Supergroup. The direction of site 34(*i*) has been corrected by vector subtraction. Sites 34(*ii*) and 35 have been thermally treated.
*Temperature in °C.

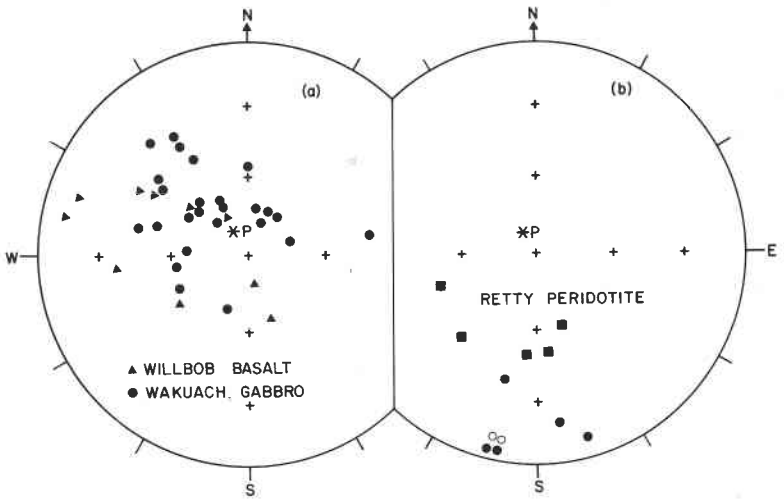


FIG. 9. Cleaned sample directions from the Willbob–Wakuach (*a*) and the Retty (*b*) rocks. In (*b*), squares are vector-corrected initially cleaned directions (15 mT) and dots are finally cleaned directions (550 °C).

related to serpentinization and the formation of magnetite shortly after intrusion, but the favoured interpretation is that the magnetization is secondary and has been acquired after folding, as in most other rocks of the Trough and other folded sills of this metamorphic grade (Ueno and Irving 1976).

Sokoman Iron Formation

The nrm directions are generally directed away

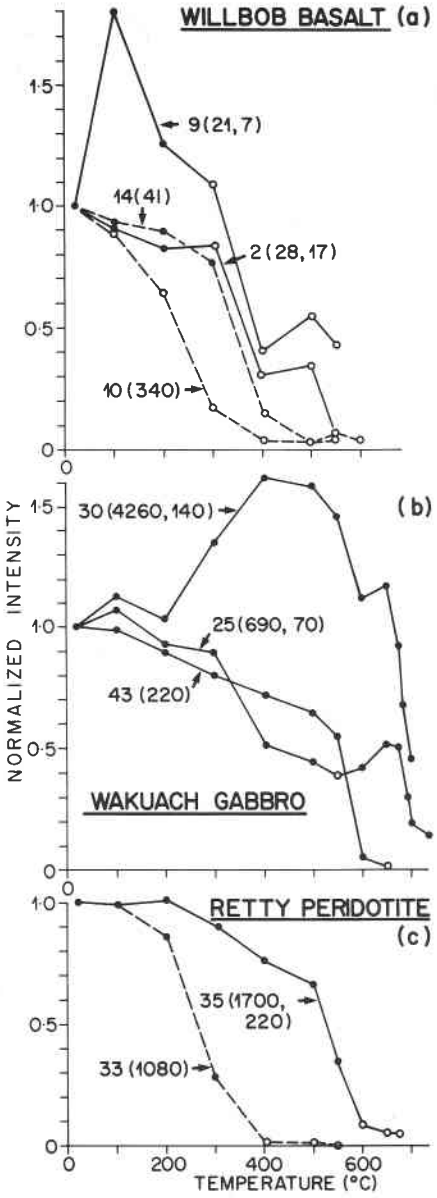


FIG. 10. Typical thermal decay curves from Willbob (*a*), Wakuach (*b*), and Retty (*c*) rocks. Unstable specimens, have dashed curves. Numbers in brackets are, first, the nrm intensity, and second, the intensity after af cleaning both given in 10^{-3} A/m. Other symbols are explained in Fig. 8.

from the present field, and have site precisions (k) ranging from 3 to 5. There is no appreciable improvement of precision during thermal cleaning. Specimens show end-points at about the Curie point of magnetite, though reasonably consistent directions can persist to the Curie point

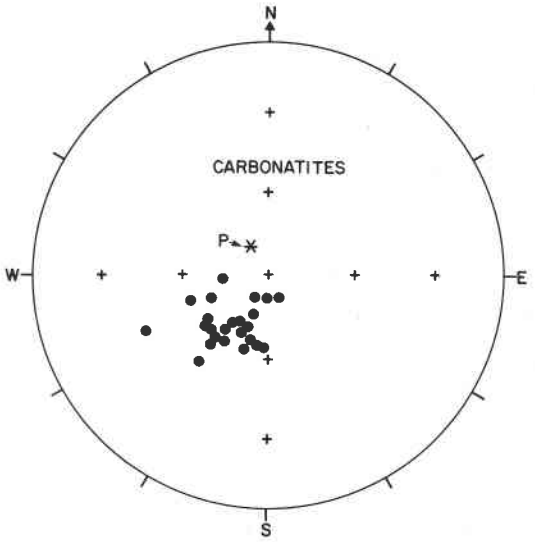


FIG. 11. Cleaned sample directions from the Castignon complex.

of hematite. Thermal demagnetization curves (Fig. 4*a*), like those of the Seward and Menihek (Fig. 4*b*), are typical of rocks that have acquired a viscous ptrm by mild heating during deep burial. These rocks (except for cherty rocks at site 42) contain 10 to 35% opaque minerals, almost all of which are magnetite and hematite, and wide scatters in direction presumably arise from magnetic interactions in these concentrated oxide phases.

Castignon Complex

Initial site precisions (k: 6 to 13) improve considerably in af cleaning fields of about 15 mT and the directions become grouped in a steep southerly direction (Table 9, Fig. 11). The average direction is given in Table 5 (entry 7). A typical af decay curve is shown in Fig. 12*b*. The thermal decay curves are variable, but often show prominent blocking temperatures between 550 and 600 °C (Fig. 12*a*). The directions remain stable up to 550 °C. Polished sections mainly contain large magnetite grains (usually about 100 μm (microns)), which probably account for the steep initial fall of the af curves. Maghemite and goethite are present as significant alteration products of the magnetite in dike 12, but are largely absent from the diatremes. This mineralogy is thus mainly indicative of low temperature oxidation. The magnetization is probably due to magnetite, and is either a primary trm produced

TABLE 9. Cleaned directions of Castignon complex

Site	UTMG	M_N	Treatment (mT)	N	$D°,I°(R)$
8	EN120451	4035	20	5	207,+63(4.97)
12	EN112493	3681	15	4(1)	194,+78(3.98)
46	EN117592	2755	20	5	232,+56(4.91)
47	EN118592	2508	20	5	222,+68(4.97)
48	EN113596	2952	20	5	212,+70(4.91)

NOTE: Symbols are explained in Tables 2 and 4.

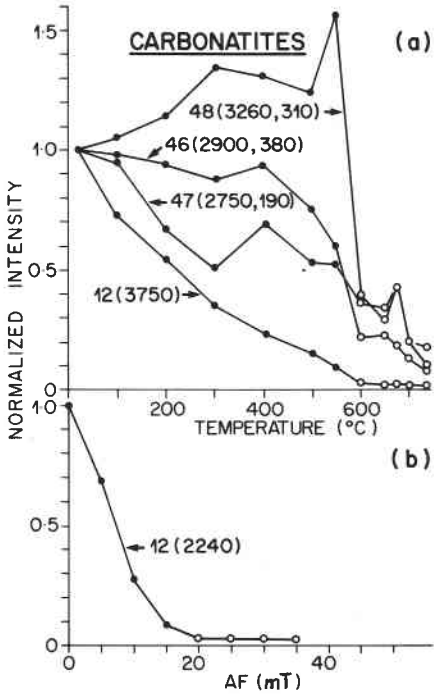


FIG. 12. Typical thermal (*a*) and af (*b*) decay curves from the Castignon complex. Symbols explained in Figs. 8 and 10.

by rapid cooling, or a crm produced by low temperature oxidation. A less stable viscous ptrm relating to the Willbob-Wakuach direction might also be present, but it could not be satisfactorily separated. There is no reason to believe that the main magnetization (*CT*) was acquired at any very considerable time after formation and it is therefore tentatively referred to the K–Ar age of 1875 Ma (see above). No tilt corrections seem to be needed because the *in situ* directions are well-grouped (Fig. 11).

High Temperature Directions

All of the carbonatites and most of the metagabbro specimens show a high-temperature direction between 600 and 735 °C. Examples from the metagabbro and carbonatite units are shown in Fig. 13. High-temperature directions were earlier found by Park (1975) to be associated with J_s–T curves with Curie points ranging from 700 to 780 °C. One J_s–T curve was obtained in the present work from site 48 carbonatite. This curve shows a distinct Curie point at about 550 °C near the Curie point of magnetite, a slight dip at 675 °C at the Curie point of hematite, and a long tail which disappears at 770 °C close to the Curie point of iron. These rocks contain many sulphides and it is possible that a reducing environment has led to the production and preservation of free iron. No iron however was detected in the polished sections.

The high-temperature directions are quite consistent in individual specimens over a range of temperatures, but vary according to the sample orientation data, so that precisions are higher

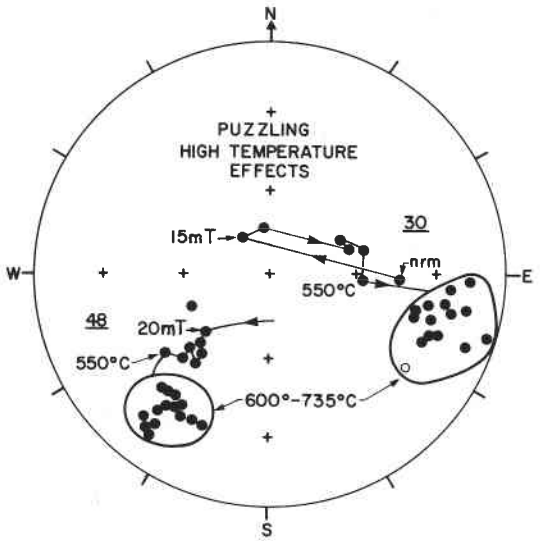


FIG. 13. Examples of high temperature directions from the Wakuach gabbro (site 30) and the Castignon complex (site 48).

when directions are referred to sample orientations. This can be explained as owing to an induced effect, which was readily apparent from comparative measurements made on Schonstedt and astatic (Roy *et al.* 1972) magnetometers. Therefore the high-temperature directions are considered unreliable, though the presence of a high-temperature component appears certain.

Proposed Chronology of Poles

Poles calculated from the magnetic directions are plotted in Fig. 14. It is not possible at present to demonstrate from internal evidence that any of the magnetizations associated with these poles are primary. In addition, although in some instances magnetizations can be separated with some confidence (*e.g.* Table 2, and separation of Seward *SC* from Seward *SA* and *SB*), in others this is not possible (*e.g.* separation of Seward *SA* from Seward *SB*), and subjective choices have had to be made. In Fig. 14 these new results are compared with the path of apparent polar wander (APW) compiled recently by Irving and McGlynn (1976). This path has been divided into tracks following the procedure of Spall (1971) and Irving and Park (1972), but the track numbering has been modified. The loop comprising tracks 3A and 3B was first drawn by Dubois (1962) and later modified by Spall (1971) and by Robertson and Fahrig (1971) who referred to it as the Logan Loop. The loop joining tracks 3B and 4A was first proposed by Spall (1971) and subsequently modified by Irving and Park (1972) and by Robertson (1973). Irving *et al.* (1972) first sketched out track 4A. Earlier than 1700 Ma, especially between 1700 and 1900 Ma, the APW path is much more tentative. An alternate configuration for track 4 given by Roy and Lapointe (1976) would extend tracks 4A and 4B much farther eastward. Tracks 4B and 4C are due to Irving and McGlynn (1976), and track 5 to Irving and Lapointe (1975), Pullaiah and Irving (1975), Irving and McGlynn (1976), and Roy and Lapointe (1976). The poles obtained in this work fall on this APW path. Parts of this curve overlap, and there is therefore some ambiguity as to which track these poles should be referred. In what follows I shall present what I consider to be the most probable age sequence of poles for the Labrador Trough based on evidence presented above. The assignments are tentative, because the results are based on a reconnaissance survey only.

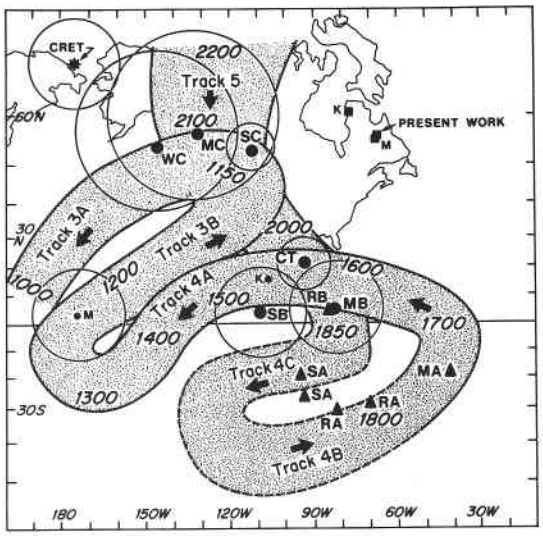


FIG. 14. Pole map. Poles of the present study are plotted as large dots (> 5 sites) and as triangles (single sites). These poles are as follows: Castignon complex (*CT*); *A* poles: Menihek Formation (*MA*), Seward Subgroup (*SA*), and Retty peridotite (*RA*); *B* poles: Menihek Formation (*MB*), Seward Subgroup (*SB*) and Retty peridotite (*RB*); *C* poles: Menihek Formation (*MC*), Seward Subgroup (*SC*), and Wakuach gabbro – Willbob basalt (*WC*). Other poles from the Circum-Ungava Geosyncline are *M* (Montagnais dikes, Seguin 1974) and *K* (komatiitic basalts, Fujiwara and Schwarz 1975). The mean Cretaceous pole (CRET) for North America is given for reference (Couillard and Irving 1974). Collecting localities are indicated. The apparent polar wander path (shaded for that part of the Canadian Shield stabilized prior to the Grenville Orogeny (from Irving and McGlynn 1976)) is shown for reference.

The poles from metasedimentary rocks (Seward *SA*, *SB*, *SC*; Menihek *MA*, *MB*, *MC*) are roughly aligned from southeast to northwest. In the northwest the *SC* and *MC* poles lie either on track 3A or on track 5. These poles have been obtained from the lowest blocking temperature magnetizations of the Seward and Menihek, and thus they probably represent the youngest of the metasedimentary magnetizations. Hence they probably lie on track 3A rather than at the 2100 Ma calibration point on track 5. The age of the other metasedimentary poles (*SB*, *SA*, *MB*, and *MA*) is uncertain, but it appears likely that *SB* and *MB* are of one general age and *SA* and *MA* of another, because the former have lower and the latter higher blocking temperatures. The age of deposition of the Menihek is considered to be 1855 $\pm$ 74 Ma (Fryer 1972). If this is indeed the age of the *MB* pole, it could belong to track 5. The pole for the Seward Subgroup likewise

roughly fits track 5. The difficulty, however, with placing *SB* and *MB* on track 5 is that the associated magnetizations were acquired after folding. It is therefore more logical to refer them to track 4A (1300 to 1700 Ma). Track 4A is populated by poles derived from magnetizations acquired during cooling following post-Hudsonian uplift. In the case of poles *SB* and *MB* from the Labrador Trough, this uplift could be represented by the mean K–Ar date of 1610 Ma (Dimroth 1970*b*, Fryer 1972), and this fits very well with the suggested chronology of track 4A. Another point in favour of this latter interpretation is the alignment of the *SB* and *MB* poles parallel to track 4A, and the fact that *SB* lies to the west of *MB*. This is logical from the uplift point of view, since the *SB* sites, being more deeply buried, would have had their magnetization reset last and their pole would fall on a younger part of the curve. This is an interesting situation in which, under conditions of deep burial, secondary magnetizations in younger strata may be acquired before those in older strata. If I am correct in assigning the *SB* and *MB* poles to track 4*A*, then the *SA* and *MA* poles are likely to be older than about 1650 Ma because they are derived from magnetizations with higher blocking temperatures. From the position of poles *SA* and *MA* on track 4B of the APW curve they too would appear to represent magnetizations that were acquired at about 1800 Ma.

The above interpretation of the *A* and *B* poles from the Seward and Menihek may also be applied to the Retty peridotite results. The less stable of the Retty magnetizations has a pole (*RB*) which lies atop the pole *MB*, and the more stable of the Retty magnetizations (*RA*) is found at two sites which have poles between the *SA* and *MA* poles. According to the APW curve the age of the *RA* pole is about 1800 Ma.

The pole *CT* for the Castignon complex lies on both track 4A and track 5. In view of its age (K–Ar, 1873 Ma) it seems reasonable to assign it tentatively to track 5. However this assignment may be in error and the magnetization could be secondary and the pole referred to track 4A.

The Willbob–Wakuach pole (*WC*) lies close to the *SC* and *MC* poles of the metasediments. The magnetizations *WC*, *SC*, and *MC* all have low blocking temperatures. It thus seems reasonable to assume that all three magnetizations were acquired at about the same time. There are several post-Hudsonian events in the Labrador Trough which might have produced these magnetizations. One event occurred in Cretaceous times. There is a slight overlap of the error circles of poles *WC* and *MC* with the Cretaceous pole (labelled CRET in Fig. 14) error circle, but the more accurately determined *SC* pole is significantly different. The correlation of these *C* poles with a Cretaceous event is therefore considered unlikely. The *C* poles are tentatively assigned to about 1150 Ma.

Other poles from the Circum-Ungava belt are shown in Fig. 14. Seguin (1974) has obtained a pole (*M*) from the as yet undated Montagnais dikes south of the sampling area (Fig. 14). This pole fits the APW curve at about 1250 Ma and may therefore belong to the Mackenzie Igneous Episode (Fahrig and Jones 1969). A pole (*K*) from volcanic rocks of the Cape Smith belt (Fujiwara and Schwarz 1975) lies west in the neighbourhood of the *B* poles. It is probably a secondary magnetization, as the authors have suggested, and related to track 4A.

The Circum-Ungava belt may have been continuous with other supracrustal rock sequences near the boundary of the Superior and Churchill Structural Provinces in Manitoba (McGlynn 1970). Rocks from the Flin Flon area may be related to this zone, and it is noteworthy that they exhibit two magnetizations that show certain similarities to the Labrador Trough magnetizations. The dominant magnetization at Flin Flon, which is probably largely the result of cooling following uplift, has poles that agree very closely with the *SB*, *MB*, and *RB* poles from the Labrador Trough. At Flin Flon, a subordinate but more stable magnetization, which is thought to be a primary trm in the ultrabasic Boundary Intrusions (dated at about 1800 Ma), has a pole near to the *A* poles from the Trough. Thus the early and later magnetizations at Flin Flon appear to have their correlatives in the Labrador Trough, indicating that the later thermal histories of both areas are broadly similar.

In summary, the poles from the Labrador Trough form a south to north trend, and lie on the APW curve of Irving and McGlynn (1976). They are also consistent with the modified version of Roy and Lapointe (1976). The oldest poles (the *A* poles) appear to be due to post-folding secondary magnetizations. These *A* magnetizations are tentatively dated at about

1800 Ma. The magnetization of the Castignon complex (*CT*) also appears to date from about this time. All three *A* magnetizations were later overprinted during post-Hudsonian uplift (at about 1600 Ma). These overprints are represented by the *B* poles of the Seward (*SB*), Menihek (*MB*), Retty (*RB*). Much later (1150 Ma) a further magnetic overprint with very low blocking temperatures produced the *C* poles of the Willbob–Wakuach (*WC*), Seward (*SC*), and Menihek (*MC*). This last magnetization is not at present identifiable with any known geological event. The above interpretations are obviously very tentative, but appear to be a logical reconstruction based on the information presently available. The trend of poles from south (oldest) to north (youngest) is determined from observed blocking temperatures; the southerly *A* poles representing magnetizations with blocking temperatures typically up to 600 °C, and the northerly *C* poles, blocking temperatures of 300 °C or less.

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